Open-source spatial planning and optimization

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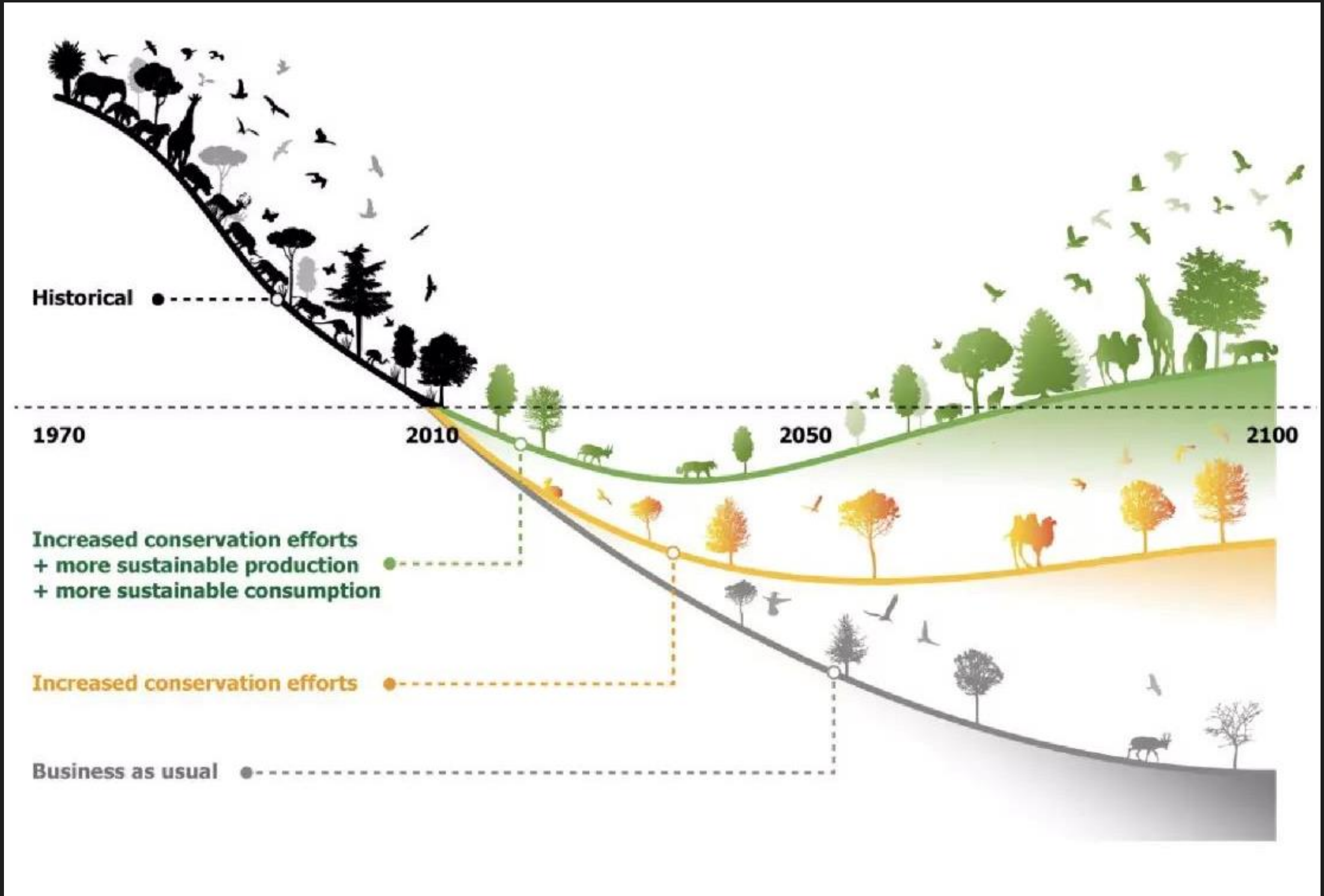
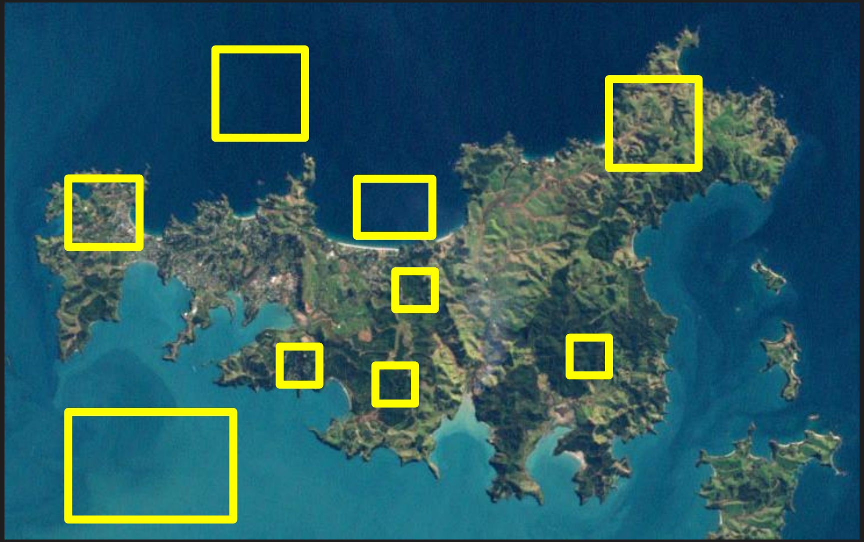
Conservation decision making



Conservation decision making is hard

- Conservation is actions in places
- Many facets of biodiversity need attention (e.g., species, habitats, ecosystem services)
- Limited resources for implementing actions
- Uncertainty in possible outcomes
- Need to find compromises among different stakeholders' values

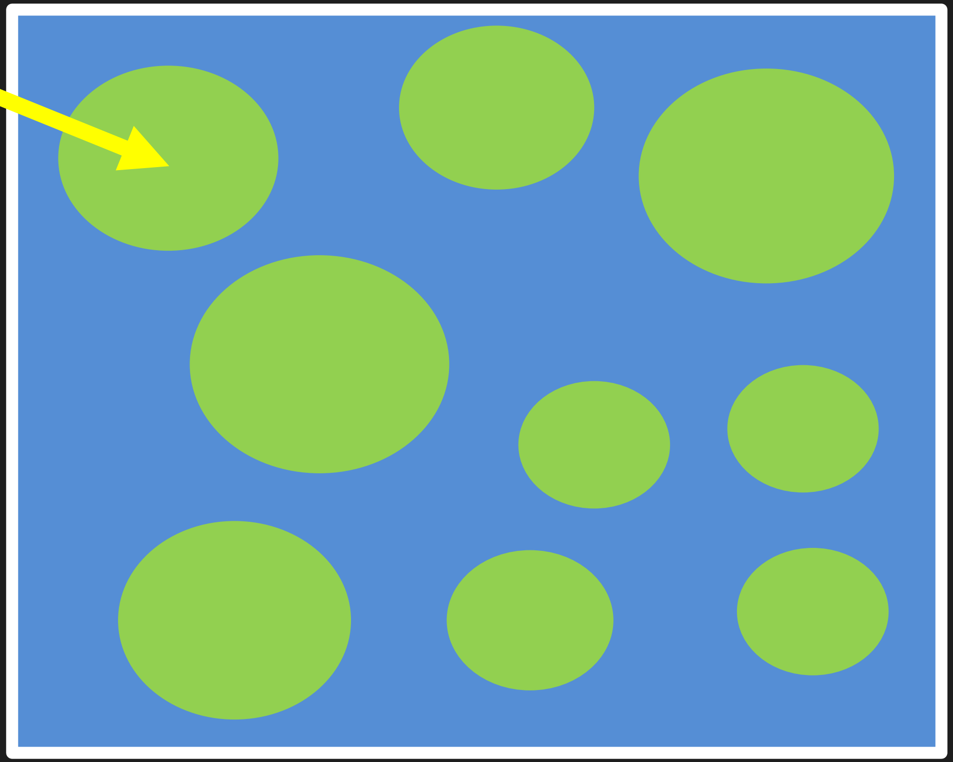


Lots of potential options for conservation efforts

Spatial optimization

Planning units

- Discrete places for conservation management
- Each planning unit is managed separately
- Commonly include land parcels, islands, spatial grid cells



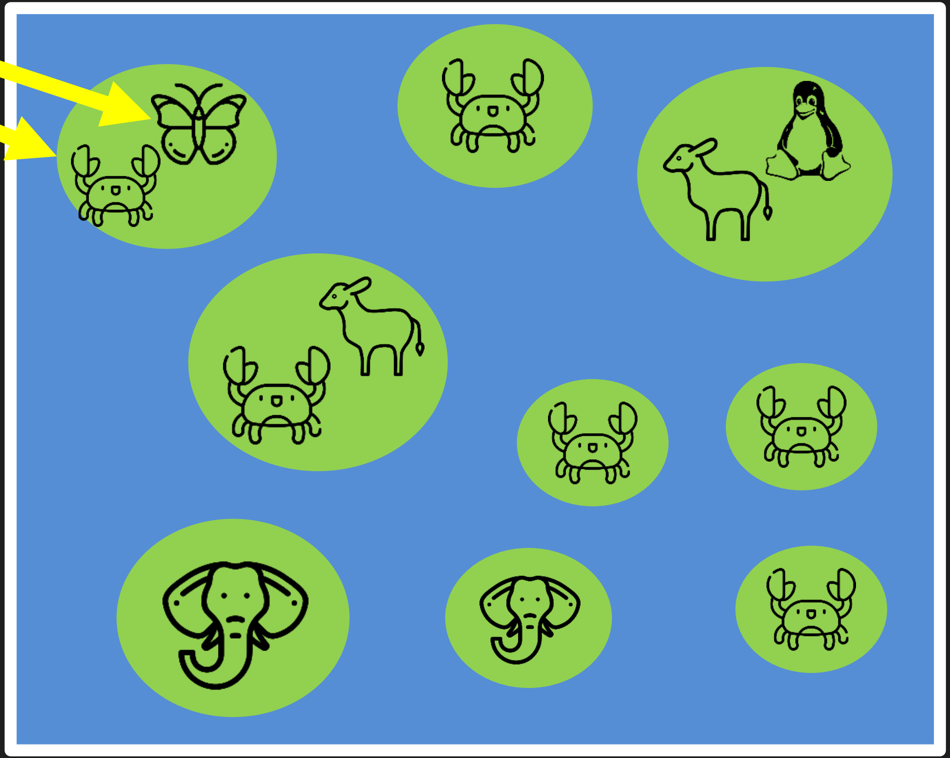
Spatial optimization

Features

- Map-able elements that we want to persist

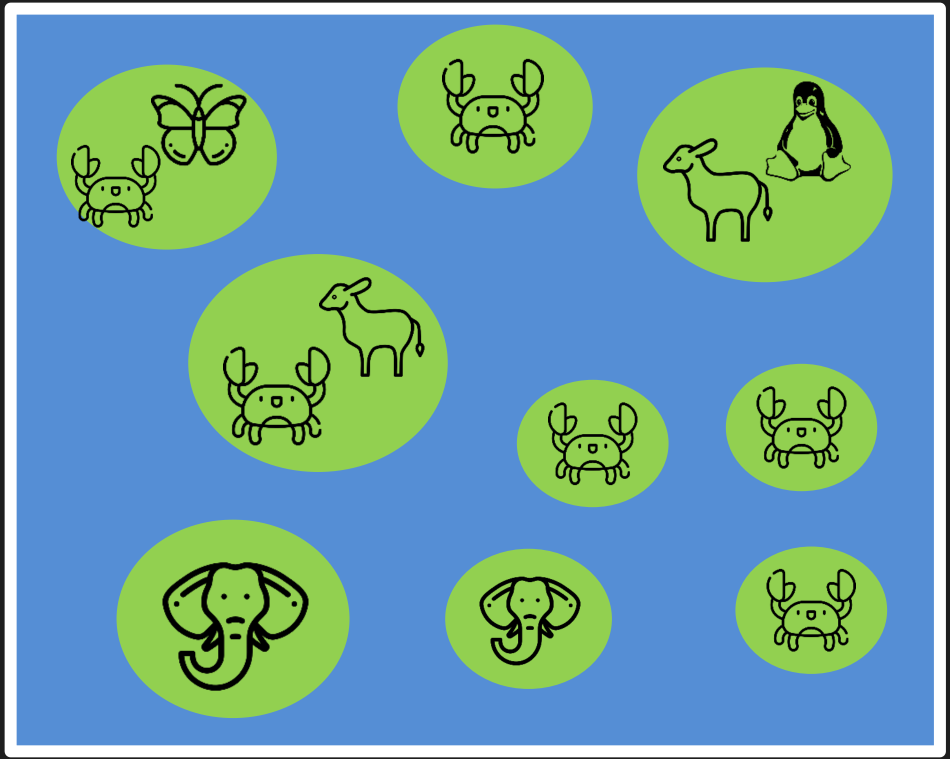
- Commonly include species, ecosystem types, ecosystem services (e.g., water provisioning, carbon sequestration)

- Target thresholds denote minimum desired amount (e.g., ≥3 crab populations)



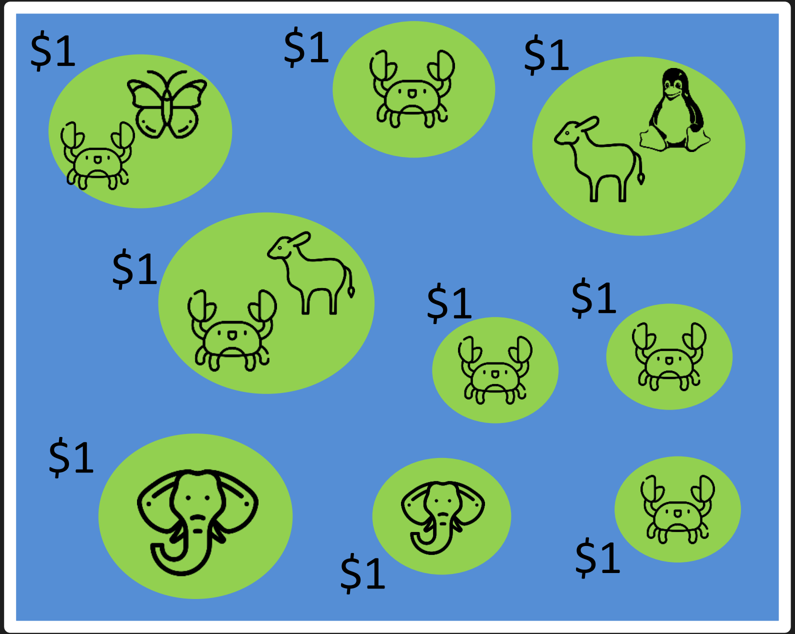
Spatial optimization

Which planning units should we manage for conservation?



Spatial optimization

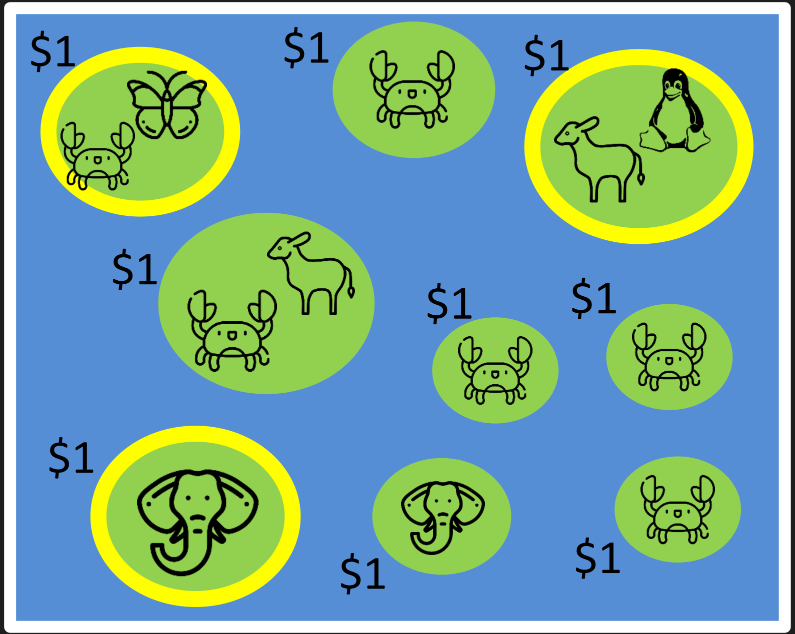
- Minimum set formulation
- Objective: min. total cost
- Constraints: meet all goals
- Cost: each island costs $1
- Targets: protect ≥1 population for each species



Spatial optimization

- Objective: min. total cost
- Constraints: meet all goals
- Targets: protect ≥1 population for each species

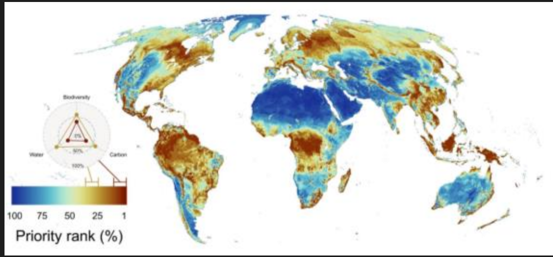
Optimal objective value = \$3



prioritizr

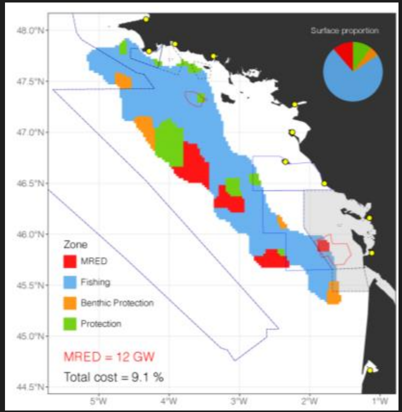
- Highly flexible
- Works with broad range of spatial and non-spatial data
- Leverages exact algorithm solvers for optimization
- Supports multiple actions and zones

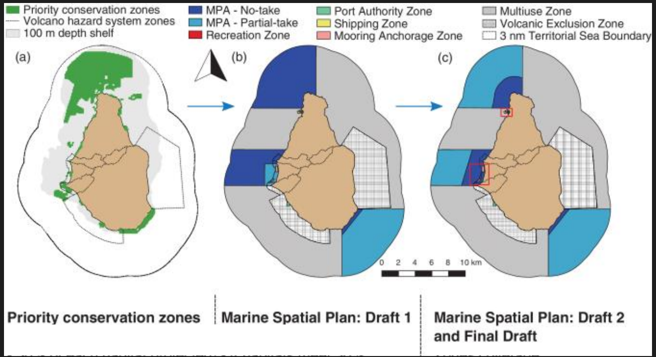
Hanson et al. 2025, https://prioritizr.net



Jung et al. 2021

Boussarie et al. 2023

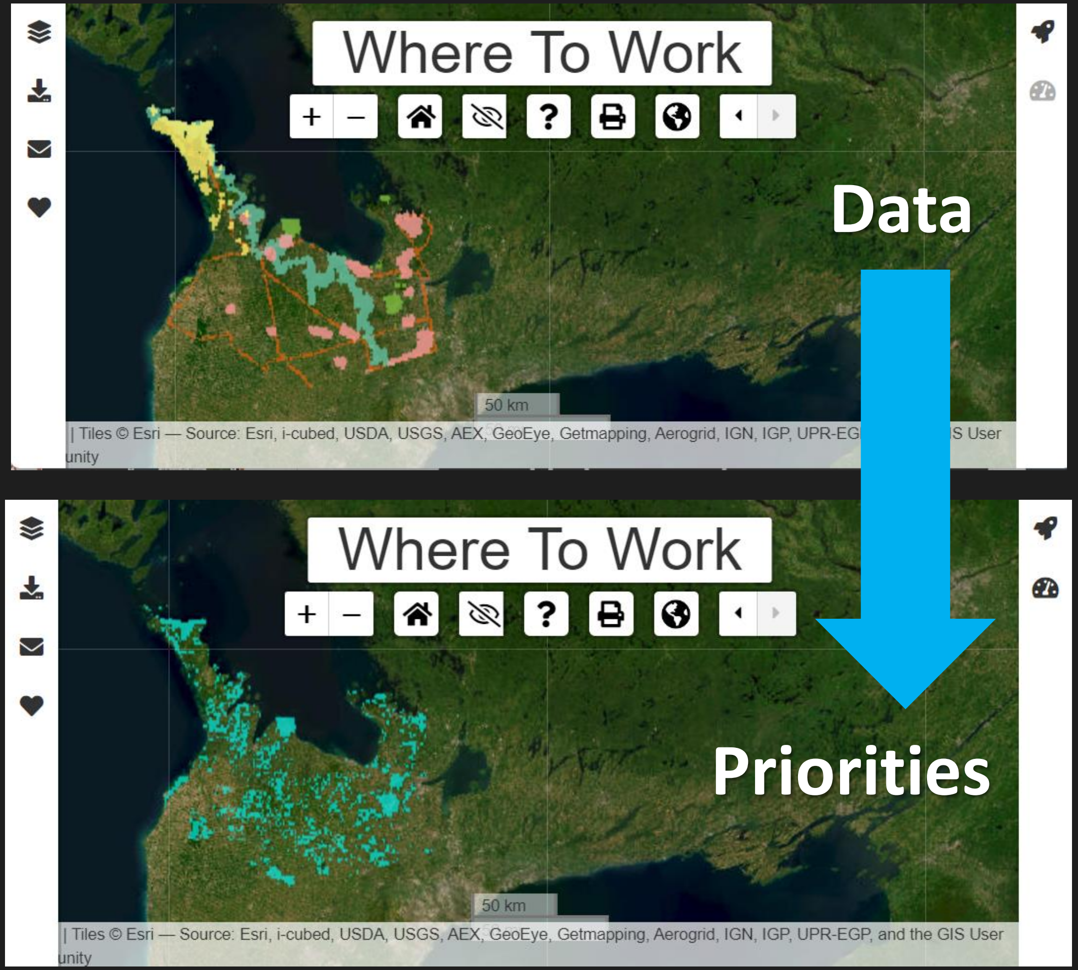




Flower et al. 2020

Where to Work

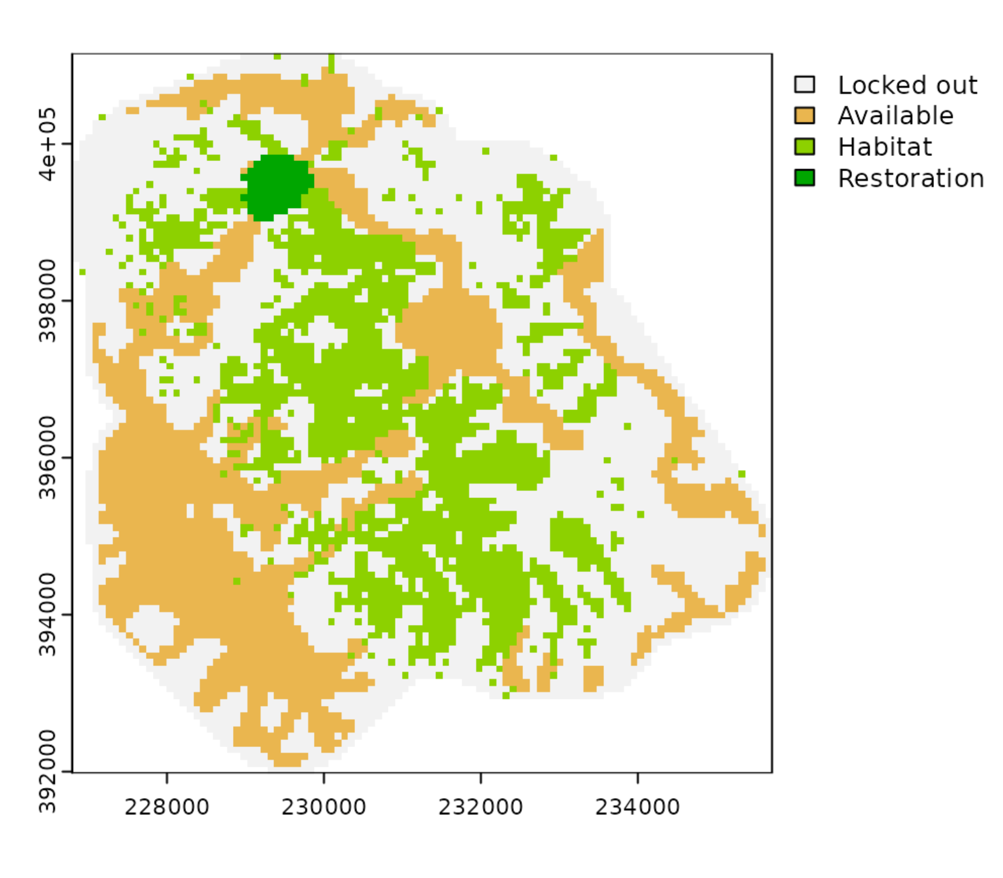
- Identifies priority areas for conservation
- Explore and resolve trade-offs between objectives
- Accessible user interface
- Developed in partnership by Nature Conservancy Canada and Carleton University



Free at https://ncc.carleton.ca

restoptr

- Identifies priority areas to restore habitat for improving connectivity
- Purpose-built for connectivity (e.g., does not support targets for multiple features)
- Supports multiple different connectivity metrics
- Prioritizations are generated based on a budget

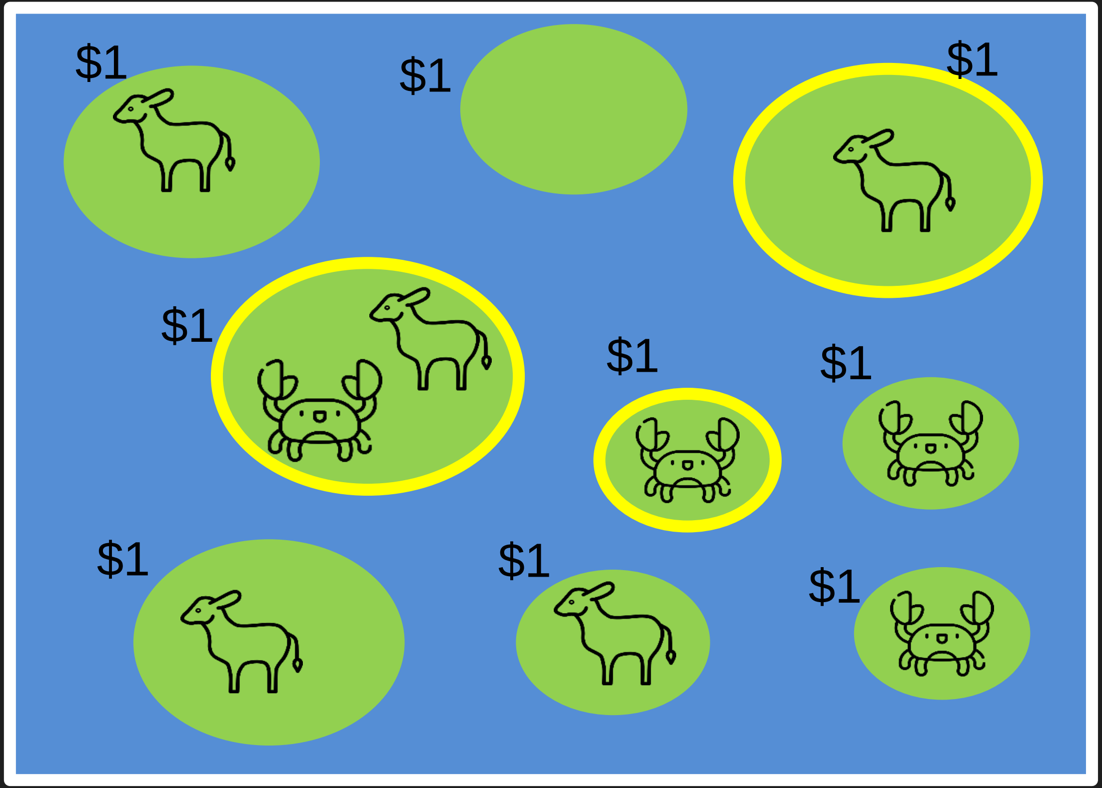


Justeau-Allaire et al. 2023, https://dimitri-justeau.github.io/restoptr

Strategies to account for global change

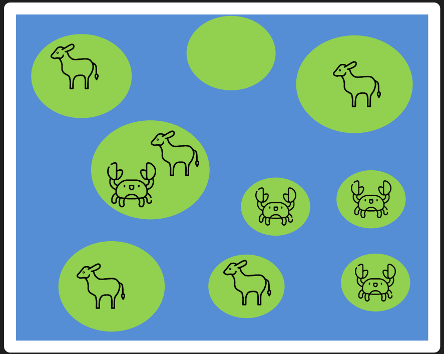
Let's consider a simple example…

- We have two species
- All islands cost the same to manage
- We want to protect at least two populations of each species



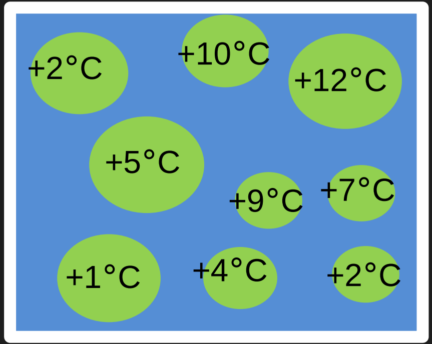
Buenafe et al. 2024, 2025

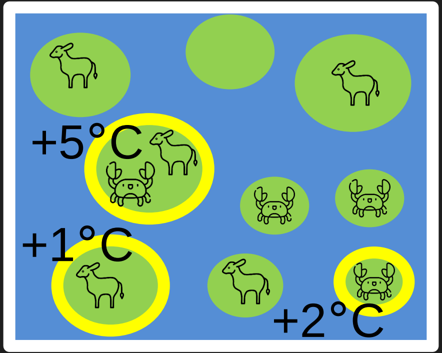
- Prioritize selection of locations with low exposure
- Useful when conservation action does NOT abate threat (e.g., climate change)
- Set cost values according to exposure (e.g., climate velocity)
- Doesn’t account for species-specific sensitivities to change



Present day

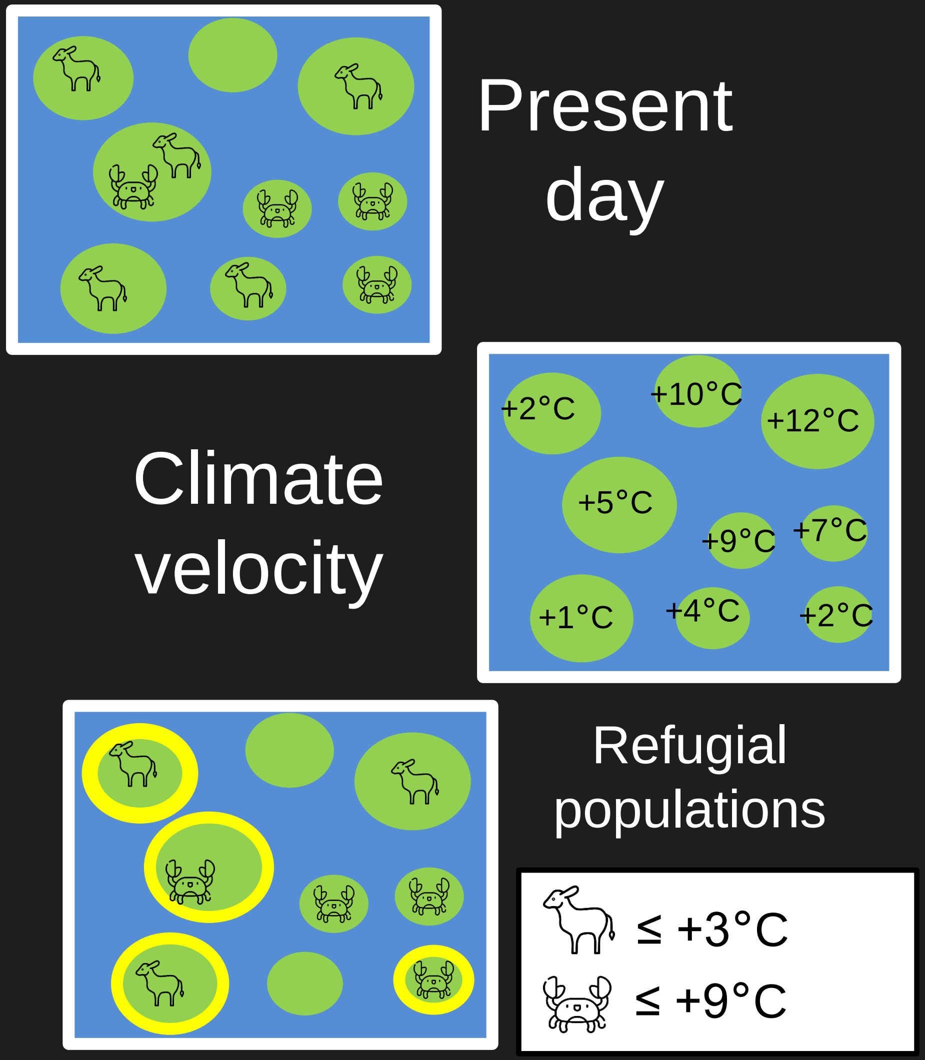
Climate velocity



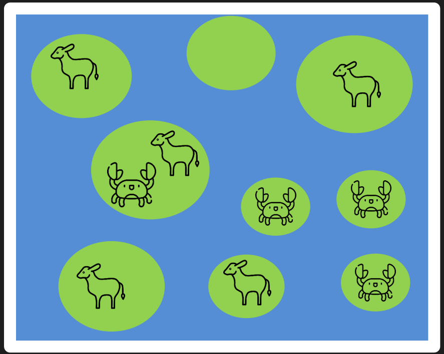


Total cost of priority areas = +8°C

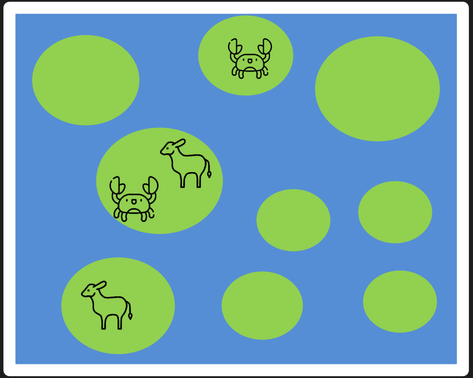
- Prioritize selection of refugia for species
- Useful when conservation action does NOT abate threat (e.g., climate change)
- Mask feature values according to exposure thresholds
- Requires threshold calibration



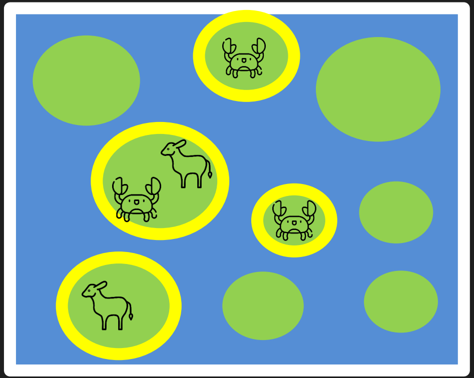
- Prioritize selection of locations based on future predictions
- Useful when conservation action does NOT abate threat (e.g., climate change)
- Set feature values according to future predictions (e.g., per statistical models)
- High data requirements



Present day

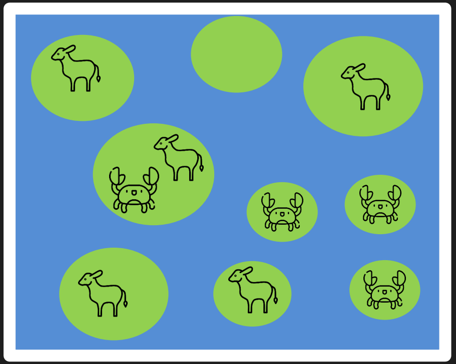


Predicted future

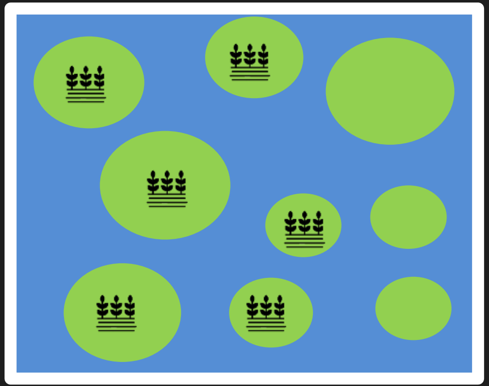


Priorities for present and future populations

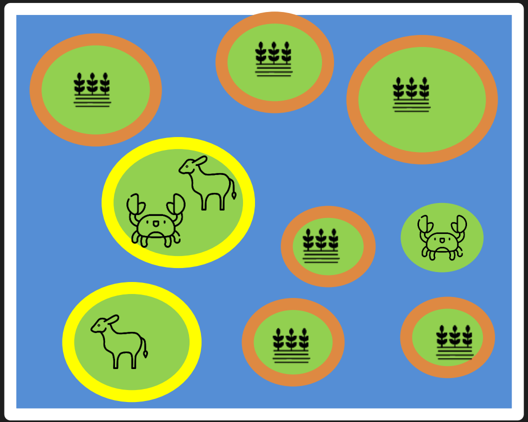
- Integrated spatial planning
- Useful when conservation action does abate threat (e.g., land use change)
- Prioritize multiple land use types with zones
- Requires objectives for each land use types
- Very high data requirements



Present day



Future land use scenario



Priorities for conservation

Priorities for agriculture

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Open source tools